RetailPulse

Retail Analytics & Intelligence Dashboard

Zidio Internship Project Report

Implementation: Microsoft Power BI Desktop

Submission date: 5 November 2026

Dataset: Online Retail II (as identified in the supplied project materials)

Scope and attribution

This report documents an existing mentor-provided Power BI dashboard and the subsequent inspection, corrections, and presentation changes carried out during the internship. It does not claim original authorship of the starter dashboard, source notebooks, machine-learning pipelines, or training outputs.

Prepared for academic/internship review. Replace this cover with your name or registration details if required by the Zidio portal.

1. Executive Summary

RetailPulse brings together retail sales, customer, geography/category, and operations analytics into a four-page Power BI Desktop report. The supplied report was opened and reviewed, then selected visual and measure issues were corrected for a clearer demonstration.

Work completed during this iteration

- Retained a separate working copy of the mentor-provided Power BI report.
- Inspected relationships and identified two active many-to-many Customer ID relationships plus an inactive country relationship.
- Created a new Average Order Value (AOV) measure based on daily totals divided by order counts.
- Adjusted revenue trend date-axis configuration.
- Rebuilt the country/category matrix using matching transaction-level fields.
- Improved customer table presentation and replaced an unreadable risk-score donut with a category distribution.
- Exported a dashboard preview PDF and uploaded the working PBIX to a personal GitHub repository, per the project handover.

Important caveat

This work is a dashboard refinement and review, not a full rebuild of the complete Streamlit and modeling stack. Saved model results and inventory simulation outputs have methodological limitations and must not be represented as independently proven business gains.

2. Problem Statement and Objectives

Retail operations data may be large and difficult to interpret when transactions, product categories, customer summaries, time series, and forecasts are viewed separately. RetailPulse aims to make those records easier to explore through a structured analytical report.

Primary objectives

- Summarize revenue, order volume, quantities, and customers.
- Identify trends over time, including monthly and hourly activity.
- Compare sales by geography and product category.
- Explore customer segmentation and lifetime-value indicators.
- Present inventory-related estimates and model-health information with appropriate caveats.

Deliverable boundary

The working artifact is a four-page Power BI Desktop (.pbix) report. A PDF export is a static preview. A public interactive URL would require authorized publishing and account permissions; its availability is not established in this report.

Success criteria for this iteration

The selected pages render without obvious broken visuals; AOV is expressed as a ratio instead of a sum; the country/category matrix no longer duplicates the same overall total for each combination; and model/forecast limitations are disclosed.

3. Data and Project Resources

The supplied project package is associated with the Online Retail II dataset and includes analytics assets beyond the Power BI report. The details below are from the internship project handover and have not been independently audited against raw source files in this report.

Resource	Reported detail
Transactional retail data	Online Retail II, two years
Daily / weekly sales	739 daily / 99 weekly records
Customer records	5,878 in the supplied project assets; the observed Power BI customer_summary table shows 4,339 rows
Segments	Six mentioned in project overview; five visible in the reviewed Page 3 chart
Additional resources	Four dashboard pages, notebooks, reports, Docker/Kubernetes and MLflow material

Observed Power BI tables

transactions, country_sales, customer_segments, customer_summary, daily_sales, forecast_30_days, inventory_recommendations, model_metrics, and product_category_sales.

Reproducibility constraint

Power Query displayed a missing local file path: Z:\Desktop\country_sales copy.csv. The report could display previously loaded data, but full refresh may require reconnecting mentor-provided CSV files using correct local paths. Refresh was avoided to preserve the working state.

4. Tool Choice and Implementation

The internship offered Streamlit or Power BI implementation routes. Power BI Desktop was selected because it offered an available local, visual workflow for inspecting and improving the supplied report, while a working Python/Streamlit environment was not prepared.

Tools actually used

- Power BI Desktop for inspecting tables, report pages, relationships, and visual settings.
- DAX for creating an Average Order Value measure.
- Power Query Editor for source inspection only; no successful refresh or dataset reconstruction is claimed.
- GitHub for storing and documenting the working Power BI report.
- Power BI PDF export for static review of the four pages.

Starter versus intern work

Mentor-provided: initial PBIX, underlying summary tables, source assets, and pre-existing forecast/churn/inventory analytical outputs.

Work performed in this iteration: backup, model inspection, a new AOV measure, chart field modifications, a corrected matrix, readability improvements, and limitations documentation. The original notebooks/models were not retrained or independently revalidated.

AOV formula

Correct AOV = DIVIDE(SUM(daily_sales[TotalSales]), SUM(daily_sales[Orders]), 0)

This is a ratio over daily summary fields. Its response to filters from unrelated tables needs further testing.

5. Page 1 — Sales Overview

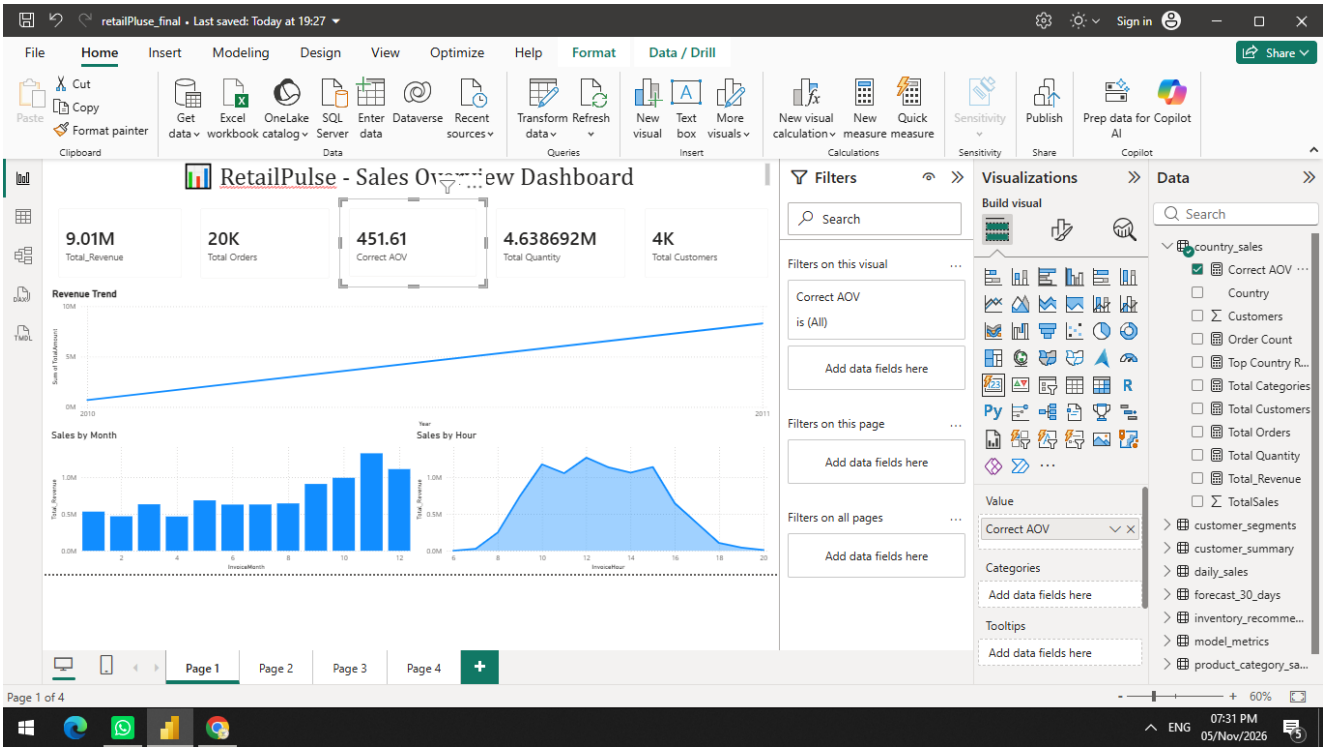


Figure 1. Sales Overview during the AOV correction workflow (screenshot captured before final currency formatting and date-axis refinement).

Key visual components

Revenue, orders, quantity, customer count, AOV, revenue trend, sales by month, and sales by hour. At the time of inspection the revenue card displayed approximately 9.01M and the orders card approximately 20K.

Correction and interpretation

A previous visual showed Sum of AOV around 135.55K. The replacement measure displayed approximately 451.61 before currency formatting. The change avoids adding AOV records together. The exact revenue and order totals behind this card should be checked against daily_sales before using the figure as an audited business metric.

6. Page 2 — Geography & Categories

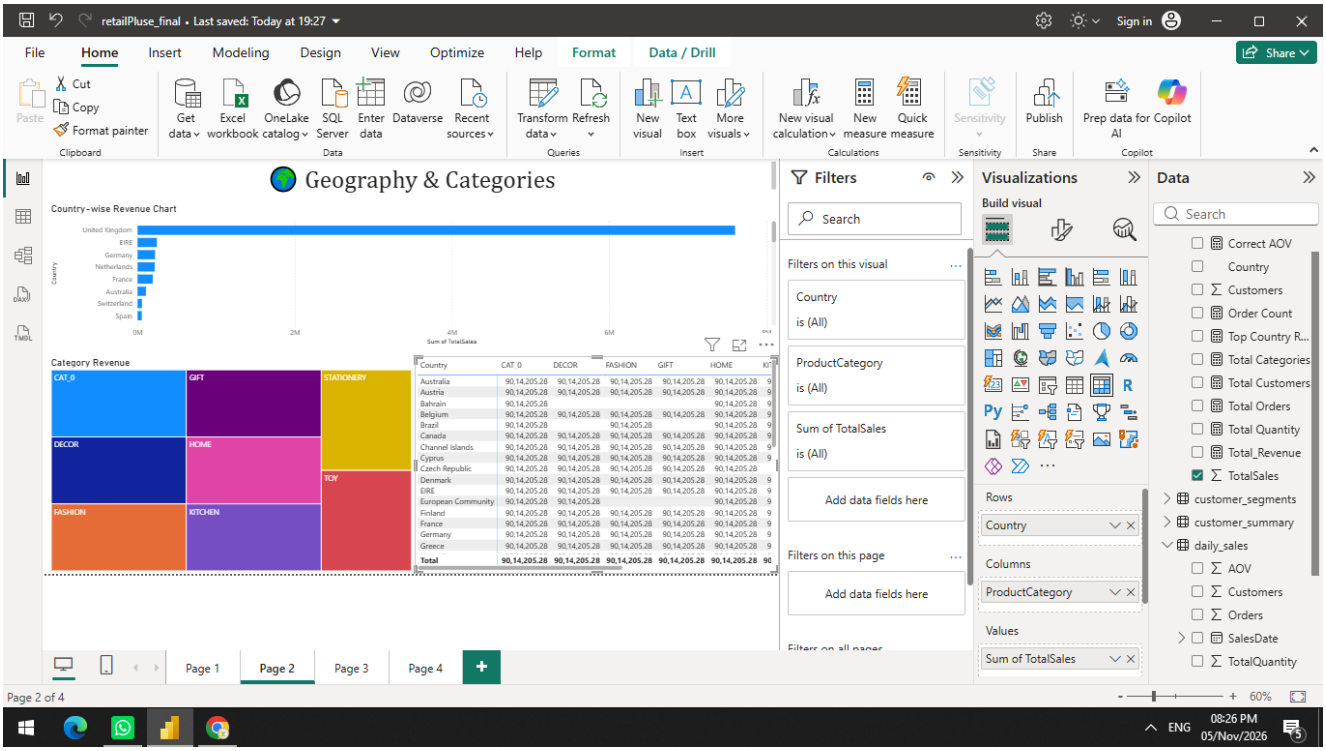


Figure 2. Geography/category page before the matrix fix: repeated 90,14,205.28 values were visible across combinations.

Identified issue

The original matrix combined Country and TotalSales from country_sales with ProductCategory from product_category_sales. Because these summary tables did not support the intended cross-filtering, the same grand total appeared in many cells.

Applied correction

The matrix was rebuilt with transactions[Country] on rows, transactions[ProductCategory] on columns, and Sum of transactions[TotalAmount] in values. The user confirmed the repeated-value issue was resolved. The measure should still be checked for returns, cancellations, and the intended sign convention.

7. Page 3 — Customer Intelligence

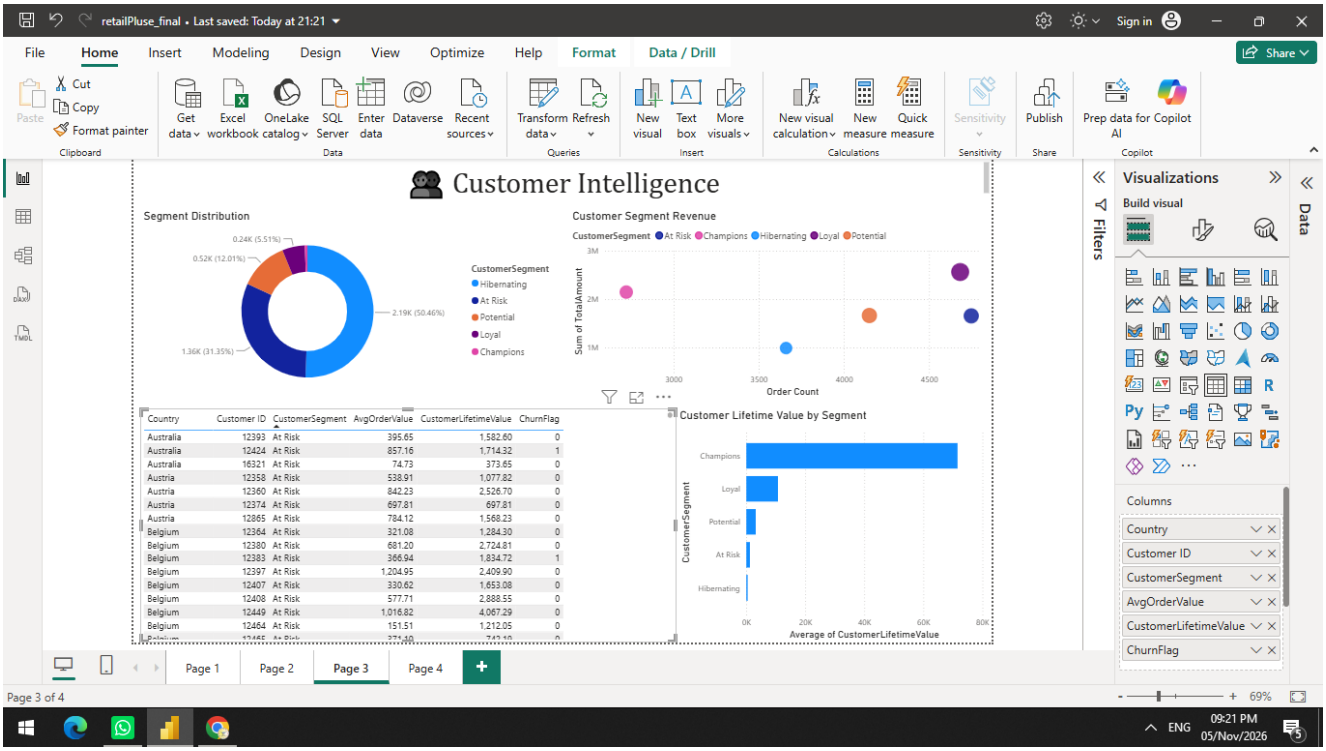


Figure 3. Customer Intelligence after table presentation adjustments.

Observed components

Segment distribution, a segment revenue scatter plot, a customer table, and a lifetime-value-by-segment chart. The customer table shows attributes such as Customer ID, CustomerSegment, AvgOrderValue, CustomerLifetimeValue, and ChurnFlag.

Interpretation and limits

Five segment names appear in the captured chart: Hibernating, At Risk, Potential, Loyal, and Champions; project-level documents refer to six. This difference needs checking for filters, missing groups, or source-table definitions. ChurnFlag should be described as a supplied indicator rather than proof of independently validated prediction.

8. Page 4 — Operations & Model Health

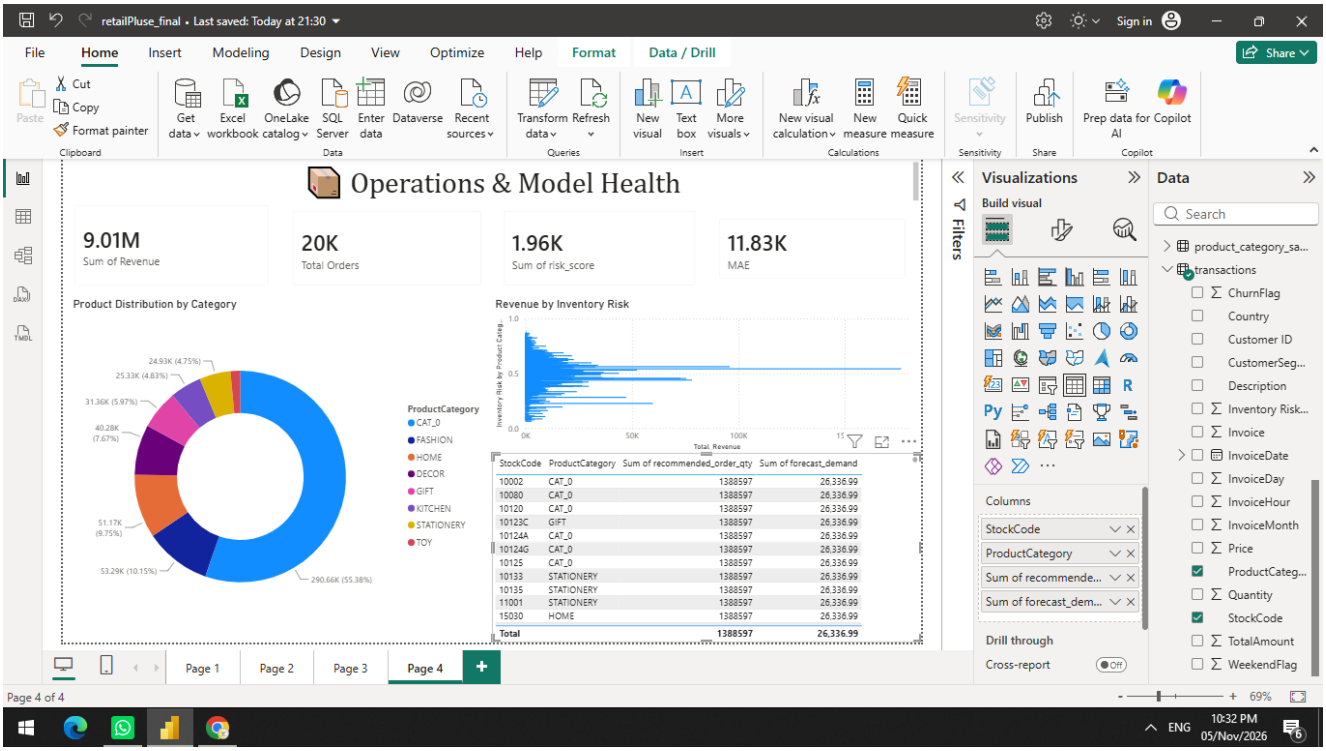


Figure 4. Operations page after replacing the unreadable numeric risk-score distribution with a category-based chart, titled Product Distribution by Category.

Observed dashboard elements

Revenue and orders cards, sum of risk_score, an MAE metric, category distribution, revenue-versus-risk display, and a stock-code table.

Outstanding quality issue

At the time of the final screenshot, the table still displayed the same recommended_order_qty and forecast_demand figures on multiple product rows. These appear to be overall aggregates being repeated rather than reliable stock-code-level values. They should be removed or corrected before relying on the table for per-item decisions.

9. Data Model and Validation Findings

Relationship inspection showed three links: transactions[Country] to country_sales[Country] (inactive), transactions[Customer ID] to customer_summary[Customer ID] (active many-to-many), and transactions[Customer ID] to customer_segments[Customer ID] (active many-to-many).

Why the relationship was not changed

Attempting many-to-one cardinality produced a Power BI blank-value warning on the one side. The original many-to-many relationship was retained rather than forcing a change under deadline pressure. The source keys, blank handling, uniqueness rules, and filtering paths need a later validation pass.

Minimum follow-up checks

- Check whether customer IDs are distinct and nonblank in each intended dimension table.
- Verify if every fact transaction has a valid associated customer key; distinguish anonymous transactions.
- Avoid bidirectional many-to-many filters unless the expected propagation is tested.
- Ensure country/category measures do not mix unconnected summaries.
- Check measure totals with sample invoices and documented treatment of credits/returns.
- Fix missing source paths before attempting an end-to-end refresh.

10. Forecasting and Inventory: Methodological Limits

The mentor-provided analytics package includes forecasting, churn, and inventory outputs. These are included here as context, not as models developed or validated during the Power BI refinement.

Forecast evaluation

A saved 3.65% MAPE for a linear stacking approach was calculated on the same four observations used to fit the stacker and therefore does not provide independent test accuracy. A saved 11-fold Prophet validation mean MAPE of approximately 23.05% was also reported in the project handover. These figures are not directly interchangeable as evidence of generalization.

Churn evaluation

The churn target was derived from recency while recency was also used among model inputs. This target leakage can make an apparent 100% ROC-AUC result misleading; it should not be quoted as reliable out-of-sample predictive accuracy.

Inventory simulation

The reported 98.9% fill rate and eight stockout days were simulated outputs. They are not measured operational improvements in an actual retail supply chain. A sum of risk scores is also not a validated risk KPI without a defined aggregation and interpretation.

11. Testing, Limitations and Next Steps

Checks completed

- Opened the supplied PBIX successfully in Power BI Desktop.
- Reviewed all four report pages and the model view.
- Created the AOV measure without a DAX syntax error and displayed it in the card.
- Adjusted the revenue date axis and rebuilt the problematic matrix.
- Improved customer table presentation and replaced the unreadable operations donut.
- Saved the working PBIX and exported a static PDF preview.

Checks not completed

- Independent verification of all KPI totals against raw invoice-level data.
- Full refresh after correcting unavailable CSV source paths.
- Reconstruction of the six-segment definition and customer relationship keys.
- Independent testing, retraining, or audit of the provided ML models.
- Confirmation of a public interactive Power BI demo URL.

Recommended next iteration

Reconnect the source files; build a clean dimensional model with deliberate filter direction; add audit-grade sales and order measures; test slicer behavior; repair stock-code-level operations fields; and perform proper temporal validation of forecast/churn models.

12. Submission Notes and Conclusion

Suggested submission contents

Artifact	Status in conversation
RetailPulse Power BI report (.pbix)	Saved and uploaded to GitHub (user confirmed)
Dashboard PDF preview	Exported locally; file location was not yet confirmed
README.md	Draft supplied; user reported GitHub upload complete
Internship report (this PDF)	Prepared from user-provided context and screenshots
4–8 minute walkthrough video	Not yet confirmed
Public interactive demo link	Not yet confirmed; subject to publishing permission

Public sharing reminder

A public repository or Power BI Publish to web link can expose embedded data. Obtain permission before publishing mentor-supplied datasets, PBIX extracts, or private information. A static PDF is not a substitute for an interactive public demo when the internship requires one.

Conclusion

This iteration converted a mentor-provided Power BI starter report into a clearer reviewable RetailPulse dashboard by correcting selected aggregations, improving the date-based sales view, rebuilding the country/category matrix, and improving operations visual readability. The strongest contribution is careful inspection and transparent correction of dashboard behavior, not claimed authorship of the broader machine-learning system.

Attribution: Initial project assets and Power BI implementation supplied by Zidio internship mentor; modifications and observations documented from the internship workflow. Screenshots are of the user’s local Power BI Desktop session.